

Math 225. if you have a function. $g(x)$ is this function the result of compute the derivative of some other function? For example if $g(x) = 2x$ then $f_1(x) = x^2$ is a function s.t $f_1'(x) = 2x = g(x)$. $f_2(x) = x^2 + 1$, $f_3(x) = x^2 - 5$.

Def. A function F is called an antiderivative of f on an interval I if $F'(x) = f(x)$ for all $x \in I$.

Thm. If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C.$$

where C is an arbitrary constant.

Corollary. If F and G are antiderivatives of f then $F - G = C$.

Examples: Find the most general antiderivative of

i). $f(x) = \sin x$

ii). $f(x) = x^n$, $n \geq 0$.

iii). $f(x) = x^{-3}$

Sln: i). $G(x) = -\cos x$ b/c $G'(x) = \frac{d}{dx}(-\cos x) = -\frac{d}{dx}(\cos x) = -(-\sin x) = \sin x$.

So, the most general antiderivative is $F(x) = -\cos x + C$.

ii). $G(x) = \frac{x^{n+1}}{n+1} + C$. then $G'(x) = \frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) + \frac{d}{dx}(C)$
 $= \frac{1}{n+1} \cdot \frac{d}{dx}(x^{n+1}) = \frac{1}{n+1} (n+1) x^{(n+1)-1}.$

$$= x^n$$

$$\text{iii). } H(x) = \frac{x^{-3+1}}{-3+1} + C = \frac{x^{-2}}{-2} \quad \text{then } H'(x) = \frac{-2}{-2} x^{-2-1} = x^{-3}$$

Table of antiderivatives formulas:

Function	Particular Antiderivative.
$\cos x$	$\sin x$
$\sin x$	$-\cos x$
$\sec^2 x$	$\tan x$
$\sec x \tan x$	$\sec x$
$x^n, n \neq -1$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{2\sqrt{x}}$	$\sqrt{x}$

Example. Find all functions g such that.

$$g'(x) = 4\sin x + \frac{2x^5 - \sqrt{x}}{x}$$

Sln: $g'(x) = 4\sin x + \frac{2x^5}{x} - \frac{\sqrt{x}}{x}; \quad \frac{x^n}{x^m} = x^{n-m}$

$$g'(x) = 4\sin x + 2x^4 - x^{-1/2}$$

Thus, we want to find an antiderivative of $4\sin x + 2x^4 - x^{-1/2}$.

Antiderivative of: $4\sin x$ is $-4\cos x$.

Antiderivative of: $2x^4$ is $2 \cdot \frac{x^5}{5}$

Antiderivative of: $x^{-1/2}$ is $\frac{x^{-1/2+1}}{-1/2+1} = \frac{x^{1/2}}{1/2} = 2\sqrt{x}$.

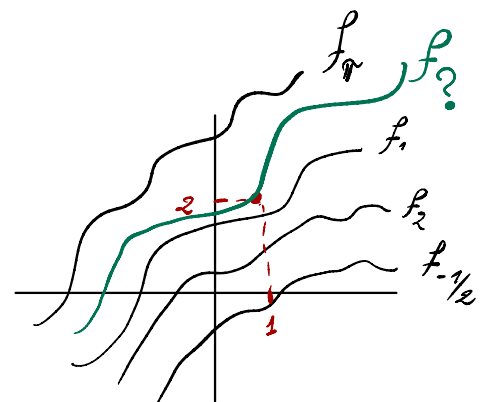
then

$$g(x) = -4\cos x + \frac{2}{5}x^5 - 2\sqrt{x} + C.$$

Exmpl: Find f if $f'(x) = x\sqrt{x}$ and $f(1) = 2$.

Sln: $f'(x) = x \cdot x^{1/2} = x^{3/2}$ so

$$f(x) = \frac{x^{3/2+1}}{3/2+1} + C = \frac{2}{5}x^{5/2} + C.$$



then $f(1)=2 = \frac{2}{5} (1)^{5/2} + C.$

$C = 2 - \frac{2}{5} = \frac{8}{5}$ then.

$$f(x) = \frac{2}{5} x^{5/2} + \frac{8}{5}$$

Example. Find f is $f''(x) = 12x^2 + 6x - 4$; $f'(0) = 4$, $f(1) = 1$.

Sln: The general derivative of $f''(x)$ is

$$f'(x) = 12 \cdot \frac{x^{2+1}}{2+1} + 6 \cdot \frac{x^{1+1}}{1+1} - 4 \frac{x^{0+1}}{0+1} + C.$$

$$f'(x) = \frac{12}{3} x^3 + \frac{6}{2} x^2 - 4x + C$$

$$f'(x) = 4x^3 + 3x^2 - 4x + C.$$

Using $f'(0) = 4$ then

$$f'(0) = C = 4$$

$$f'(x) = 4x^3 + 3x^2 - 4x + 4$$

then

$$f(x) = \frac{4x^4}{4} + \frac{3x^3}{3} - \frac{4x^2}{2} + 4x + D$$

$$f(x) = x^4 + x^3 - 2x^2 + 4x + D.$$

then $f(1) = 1$

$$1 = f(1) = 1^4 + 1^3 - 2(1)^2 + 4(1) + D.$$

$$1 = 2 - 2 + 4 + D$$

$$D = -3$$

So,

$$f(x) = x^4 + x^3 - 2x^2 + 4x - 3.$$

Linear Motion:

Example: A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6 \text{ cm/s}$

and its initial displacement is $s(0) = 9 \text{ cm}$. Find its position function.

Sln: $s(t) \Rightarrow s'(t) = v(t) \Rightarrow s''(t) = v'(t) = \underline{a(t)}$.

Since $v'(t) = a(t) = 6t + 4$ then

$$v(t) = \frac{6t^2}{2} + 4t + C.$$

$$v(t) = 3t^2 + 4t + C.$$

$$-6 = v(0) = 0 + 0 + C \Rightarrow C = -6.$$

$$\boxed{v(t) = 3t^2 + 4t - 6.}$$

Since $s'(t) = v(t)$ then, the antiderivative of $v(t)$ is

$$s(t) = t^3 + 2t^2 - 6t + D.$$

$$9 = s(0) = 0 + 0 + 0 + D \Rightarrow D = 9.$$

$$\boxed{s(t) = t^3 + 2t^2 - 6t + 9}$$

Example: A ball is thrown upward with a speed of 48 ft/s from the edge of a cliff, 432 ft above the ground. Find its height above the ground t second later. When does it reach its maximum height? When does it hit the ground?

problem for you!

$$a(t) = -32.$$

(Appendix E: Sigma Notation and 4.1 Areas and Distances)

Def. If $a_m, a_{m+1}, \dots, a_n$ are real numbers and m and n are integers numbers such that $m \leq n$ then

$$\sum_{i=m}^n a_i = a_m + a_{m+1} + a_{m+2} + \dots + a_n.$$

Notation: this tells us to end with $i=n$.
 this tells us to add a_i .
 this tells us to start with $i=m$.

$$\sum_{i=m}^n a_i$$

Examples: i). $\sum_{i=1}^4 i^2 = 1^2 + 2^2 + 3^2 + 4^2$

ii). $\sum_{i=3}^n i = 3 + 4 + 5 + \dots + (n-1) + n$

iii). $\sum_{j=0}^5 2^j = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 = 63$

iv). $\sum_{k=1}^n \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}$

v). $\sum_{i=1}^3 \frac{i-1}{i^2+3} = \frac{1-1}{1^2+3} + \frac{2-1}{2^2+3} + \frac{3-1}{3^2+3} = \frac{13}{42}$

Example: write the sum $2^3 + 3^3 + \dots + n^3$ in sigma notation.

Sln: There is no unique way of writing a sum in sigma notation

$$\begin{aligned} 2^3 + 3^3 + \dots + n^3 &= \sum_{i=2}^n i^3 \\ &= \sum_{j=1}^n (j+1)^3 \\ &= \sum_{k=0}^n (k+2)^3 \end{aligned}$$

Thm. If $c \in \mathbb{R}$ is any constant, then

i). $\sum_{i=1}^n c \cdot a_i = c \cdot \sum_{i=1}^n a_i$ ii). $\sum_{i=1}^n (a_i \pm b_i) = \sum_{i=1}^n a_i \pm \sum_{i=1}^n b_i$

Rmk: find $\sum_{i=1}^n 1 = \underbrace{1+1+1+\dots+1}_{n\text{-terms}} = n$

Thm: Let c be a constant and n a positive integer. Then.

a). $\sum_{i=1}^n 1 = n$

$$b). \sum_{i=1}^n i = \frac{n(n+1)}{2} \quad \text{if } \sum_{i=1}^5 i = \frac{5(5+1)}{2} = \frac{5(6)}{2} = 15.$$

$$c). \sum_{i=1}^n c = n \cdot c.$$

$$d). \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

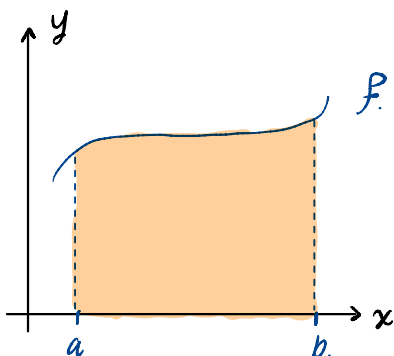
$$e). \sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Example. Evaluate $\sum_{i=1}^n i(4i^2 - 3i)$.

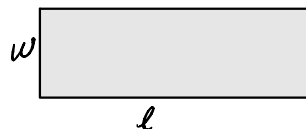
Sln:

$$\begin{aligned} \sum_{i=1}^n i(4i^2 - 3i) &= \sum_{i=1}^n (4i^3 - 3i^2) \\ &= \sum_{i=1}^n 4i^3 - \sum_{i=1}^n 3i^2 \\ &= 4 \cdot \sum_{i=1}^n i^3 - 3 \sum_{i=1}^n i^2 \\ &= 4 \left\{ \left(\frac{n(n+1)}{2} \right)^2 \right\} - 3 \left\{ \frac{n(n+1)(2n+1)}{6} \right\} \\ &= n^2(n+1)^2 - \frac{1}{2} n(n+1)(2n+1) \\ &= n(n+1) \left\{ n(n+1) - \frac{2n+1}{2} \right\} \\ &= n(n+1) \left(\frac{2n^2 + 2n - 2n - 1}{2} \right) \\ &= \frac{n(n+1)(2n^2 - 1)}{2}. \end{aligned}$$

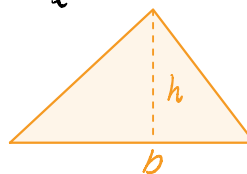
The Area Problem. We know from Math 224 that if $y=f(x)$ the the graphic is.



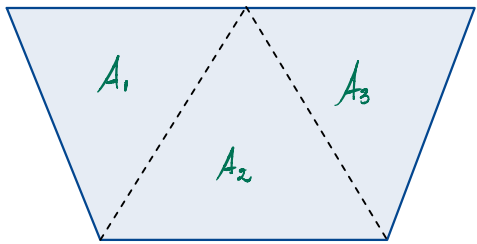
We know that



$$A = l \cdot w$$

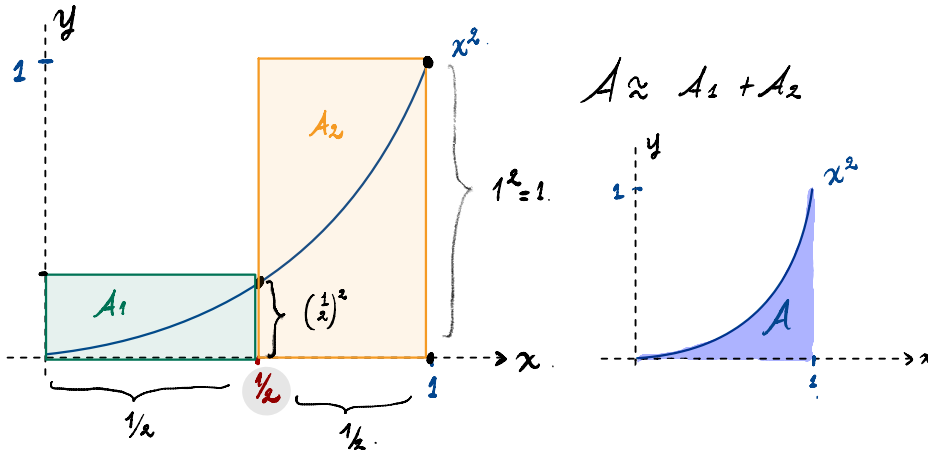


$$A = \frac{1}{2} b \cdot h.$$



$$A = A_1 + A_2 + A_3.$$

Example: Use rectangles to estimate the area under the parabola $y = x^2$ from $x=0$ to $x=1$.



$$A \approx A_1 + A_2$$

$$A = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)(1)^2 = \frac{1}{8} + \frac{1}{2} = \frac{1+4}{8} = \frac{5}{8}.$$

For you! to approximate the area using three rectangles.